

From Data to Solutions



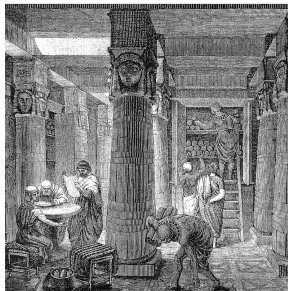
RAG foundations

Weekend 5 · Retrieval Augmented Generation

Friday 14 August 2026

Carlos Cotrini

Alexandria, about 250 BCE



- A scroll has no title page. To learn what it is, you unroll it.

Imagined by O. Von Corven, 19th century.

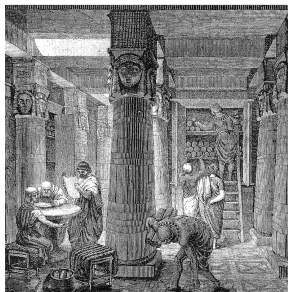
About 250 BCE

121 CE

1230

1895

Alexandria, about 250 BCE



- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**

Imagined by O. Von Corven, 19th century.

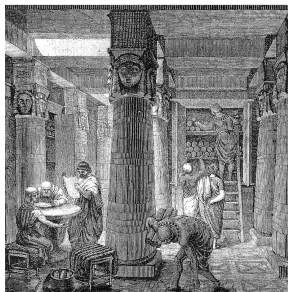
About 250 BCE

121 CE

1230

1895

Alexandria, about 250 BCE



Imagined by O. Von Corven, 19th century.

- A scroll has no title page. To learn what it is, you unroll it.
- A scholar arrives wanting the works of one author.
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.

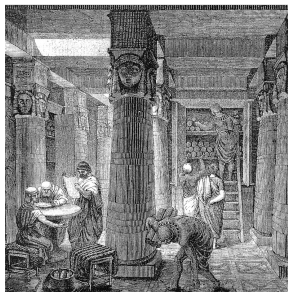
About 250 BCE

121 CE

1230

1895

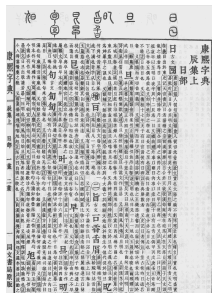
Alexandria, about 250 BCE



Imagined by O. Von Corven, 19th century.

- A scroll has no title page. To learn what it is, you unroll it.
- A scholar arrives wanting the works of one author.
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.
- You read the entries until you know which of **the scrolls** to unroll.

China, 121 CE



- A writing system with no alphabet has no order to sort by.

A page of the Kangxi Dictionary, 1716.

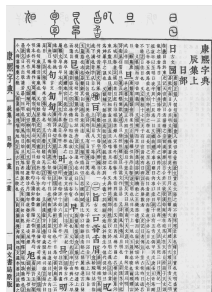
About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.

A page of the Kangxi Dictionary, 1716.

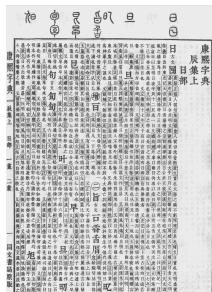
About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.
- Follow the address and you reach **the entry: what it means, how it is said**.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.

Hugh of Saint-Cher. Treviso, 1352.



Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**

Hugh of Saint-Cher. Treviso, 1352.



Paris, 1230



Hugh of Saint-Cher. Treviso, 1352.

- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**



Paris, 1230



Hugh of Saint-Cher. Treviso, 1352.

- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**
- The address sends you to **the passages themselves.** We still cite that way.



Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.



Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**

About 250 BCE

121 CE

1230

1895

Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.

Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.
- The cards told you where to find **the books and the articles**. Over 1,500 questions a year by 1912.

About 250 BCE

121 CE

1230

1895

The same three things, four times

Query	Keys	Values

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry
Paris, 1230	Every passage using a word	Book, chapter, letter A to G	The passages

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry
Paris, 1230	Every passage using a word	Book, chapter, letter A to G	The passages
Brussels, 1895	A question sent by post	Fifteen million index cards	The books and articles

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry
Paris, 1230	Every passage using a word	Book, chapter, letter A to G	The passages
Brussels, 1895	A question sent by post	Fifteen million index cards	The books and articles
A company, 2026	A question in plain words	?	The company's own documents

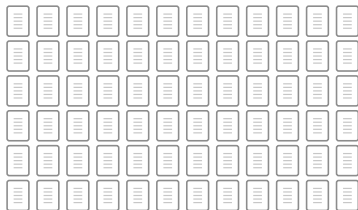
The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry
Paris, 1230	Every passage using a word	Book, chapter, letter A to G	The passages
Brussels, 1895	A question sent by post	Fifteen million index cards	The books and articles
A company, 2026	A question in plain words	?	The company's own documents

This hour builds the missing column, so that a language model can answer from those documents.

How do we do retrieval with LLMs?

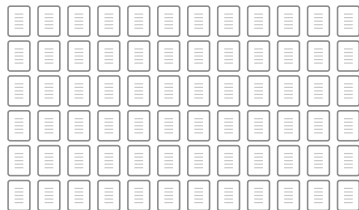
“What did we agree
with the rope supplier
about late delivery?”



all of our documentation

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



all of our documentation

all of it, at once

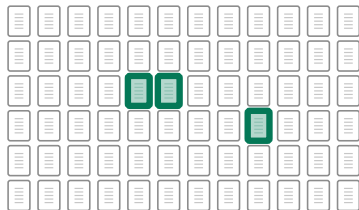


it does not fit

LLM

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



all of our documentation



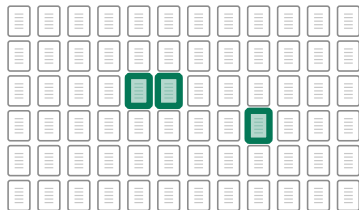
only the few
that answer it



LLM

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



all of our documentation

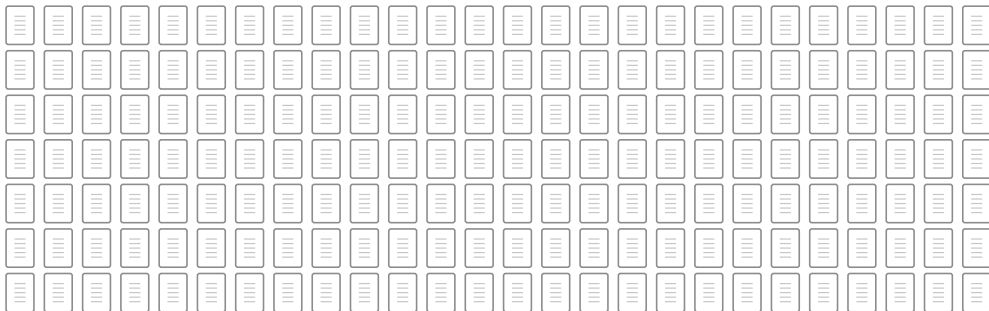


an answer, from the
documents it was shown

Finding those few, reliably, is what this hour is about.

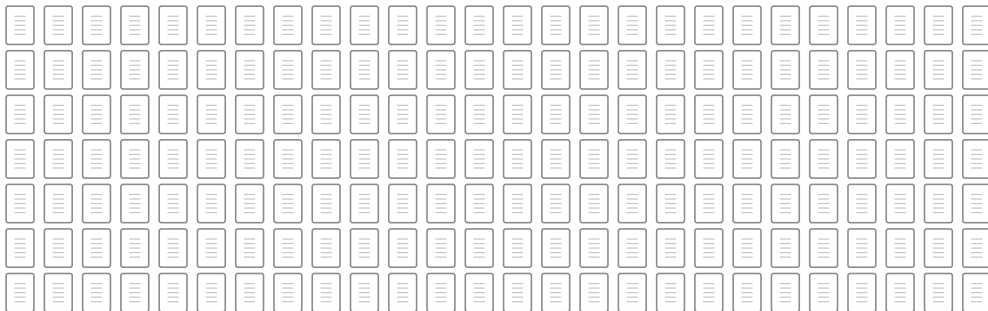
One company's own documents

Finding the few documents that answer a question



One company's own documents

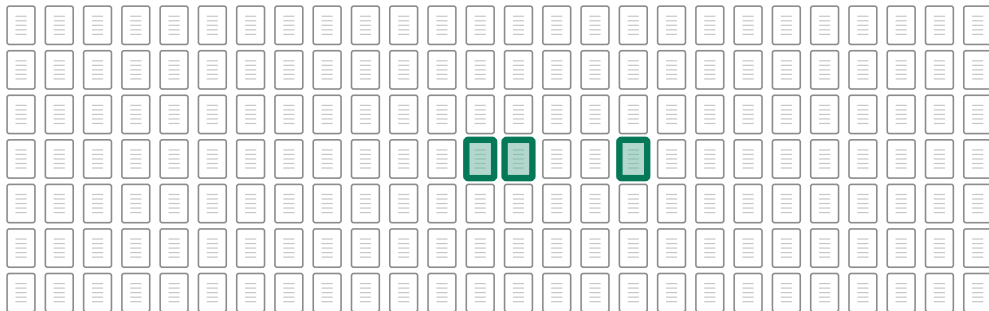
Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

One company's own documents

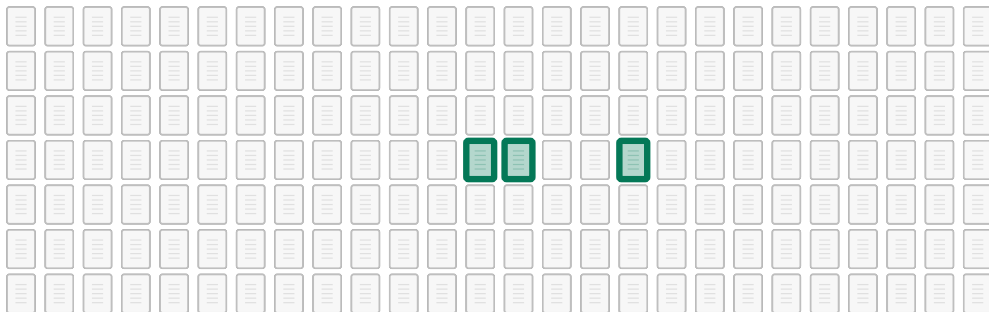
Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

One company's own documents

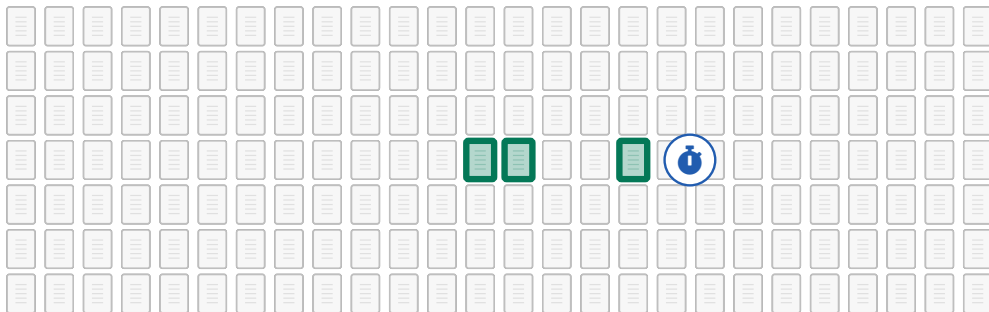
Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

One company's own documents

Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

The pub quiz game

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

Fifty documents

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

Fifty documents Six questions, at your own pace

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

Fifty documents

Six questions, at your own pace

Find the card that answers it

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

eth-fdd-fs26.github.io/FDD-Page/document-pile



Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

Fifty documents

Six questions, at your own pace

Find the card that answers it

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

5 Training record

6 Supplier contract clause

7 Weather and closure rule

8 Technical data sheet

9 Lost property record

10 Service notice

11 Weather and closure rule

12 Maintenance log

13 Training record

14 Service notice

15 Staff handbook section

16 Lost property record

17 Supplier contract clause

18 Restaurant notice

19 Lost property record

20 Incident report

twenty of the fifty shown, in no order

Motivation

The six times you measured

eth-fdd-fs26.github.io/FDD-Page/document-pile

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

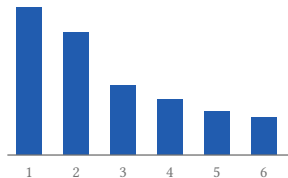
Supplier contract clause

Technical data sheet

Motivation

The six times you measured

your six rounds, on your own screen



illustrative: everyone's differs

eth-fdd-fs26.github.io/FDD-Page/document-pile

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

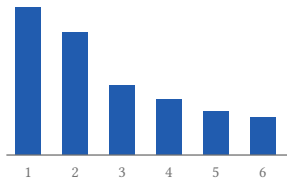
Supplier contract clause

Technical data sheet

Motivation

The six times you measured

your six rounds, on your own screen



illustrative: everyone's differs

___ of ___ hands: my last round beat my first

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

Supplier contract clause

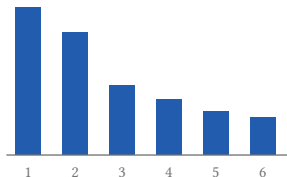
Technical data sheet

eth-fdd-fs26.github.io/FDD-Page/document-pile

Motivation

The six times you measured

your six rounds, on your own screen



illustrative: everyone's differs

___ of ___ hands: my last round beat my first

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

Supplier contract clause

Technical data sheet



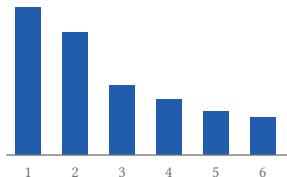
eth-fdd-fs26.github.io/FDD-Page/document-pile



Motivation

The six times you measured

your six rounds, on your own screen



illustrative: everyone's differs

___ of ___ hands: my last round beat my first

Restaurant notice	Maintenance log
Weather and closure rule	Service notice
Supplier contract clause	Technical data sheet



Incident report [][][][][]	Supplier contract clause [][][][][]
Technical data sheet [][][][][]	Restaurant notice [][][][][]
Maintenance log [][][][][]	Lost property record [][][][][]
Weather and closure rule [][][][][]	Training record [][][][][]
Staff handbook section [][][][][]	Service notice [][][][][]

eth-fdd-fs26.github.io/FDD-Page/document-pile

Queries, keys and values

VALUES

cards, not chapters



Queries, keys and values

QUERIES

How much do we get back for each extra day a consignment is overdue?

VALUES

cards, not chapters



Queries, keys and values

QUERIES

How much do we get
back for each extra
day a consignment is
overdue?

KEYS

four of the ten keys shown

Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

VALUES

cards, not chapters



Queries, keys and values

QUERIES

How much do we get
back for each extra
day a consignment is
overdue?



KEYS

four of the ten keys shown

Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

VALUES

cards, not chapters



Queries, keys and values

QUERIES

How much do we get
back for each extra
day a consignment is
overdue?

KEYS

four of the ten keys shown

Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

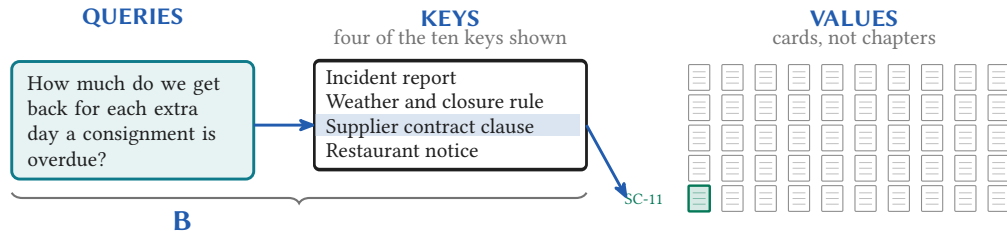
VALUES

cards, not chapters

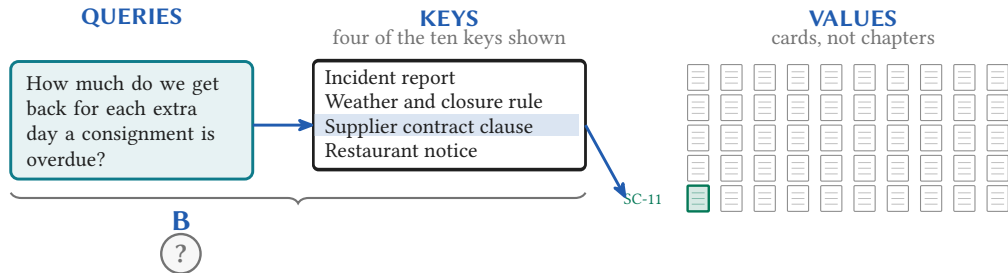


SC-11

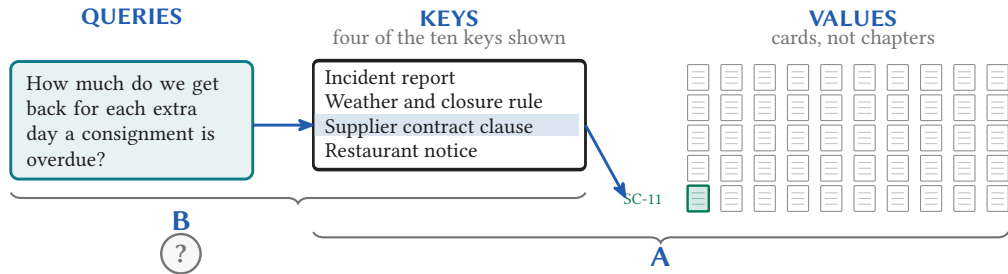
Queries, keys and values



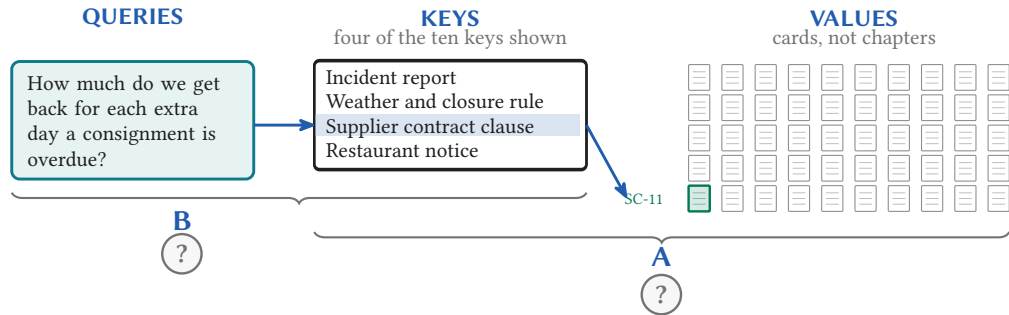
Queries, keys and values



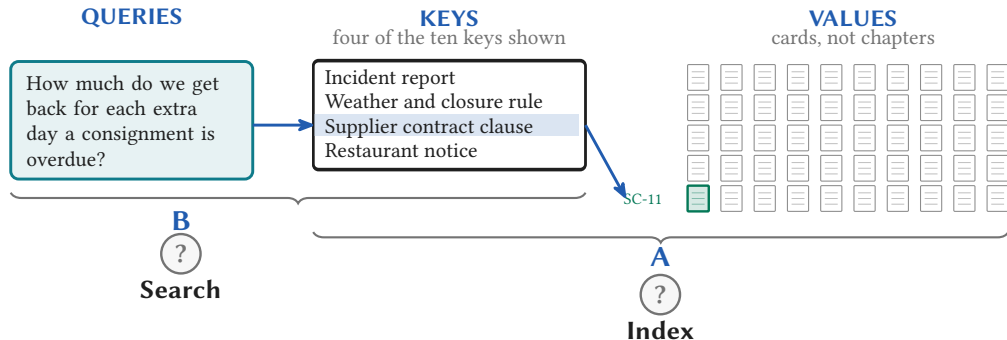
Queries, keys and values



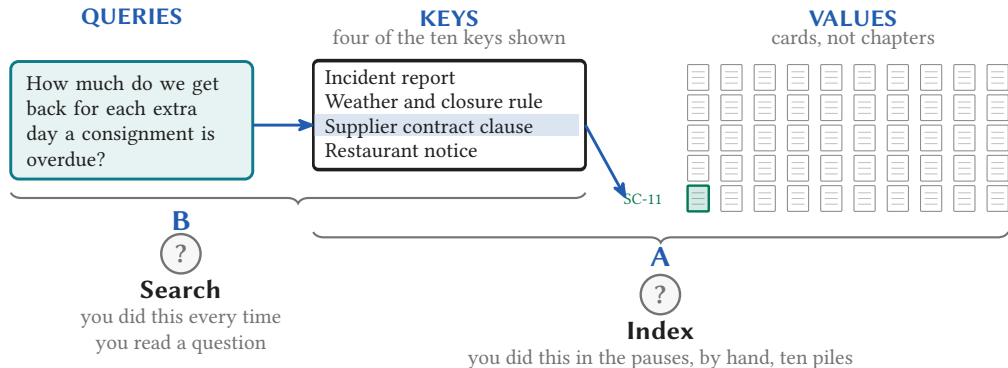
Queries, keys and values



Queries, keys and values



Queries, keys and values



The yogurt cup game



[FDD-Page/yogurt-cups](#)

The yogurt cup game



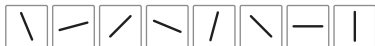
[FDD-Page/yogurt-cups](https://fdd-page.github.io/yogurt-cups)



sender



receiver



The yogurt cup game

- Both see every picture



[FDD-Page/yogurt-cups](https://fdd-page.github.io/yogurt-cups)



sender



receiver



the same eight
of the twenty four



The yogurt cup game

- Both see every picture
- One sees the target



[FDD-Page/yogurt-cups](https://fdd-page.github.io/yogurt-cups)



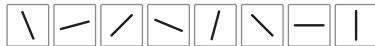
sender



receiver



the same eight
of the twenty four

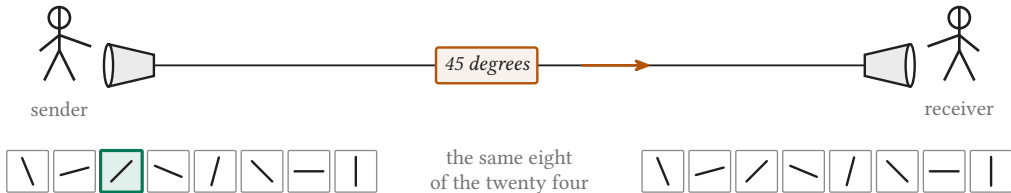


The yogurt cup game

- Both see every picture
- One sees the target
- Five words through the cup



FDD-Page/yogurt-cups

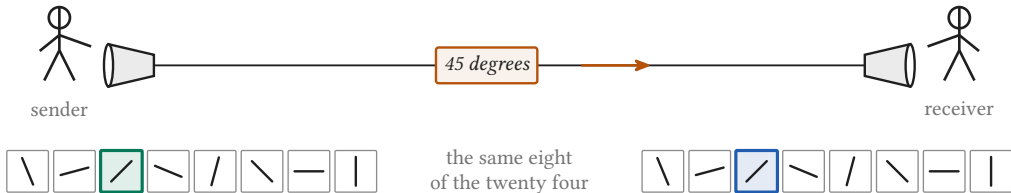


The yogurt cup game

- Both see every picture
- One sees the target
- Five words through the cup
- The partner picks one



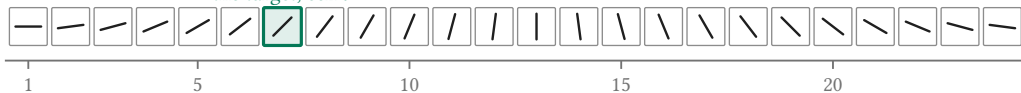
FDD-Page/yogurt-cups



Partial credit

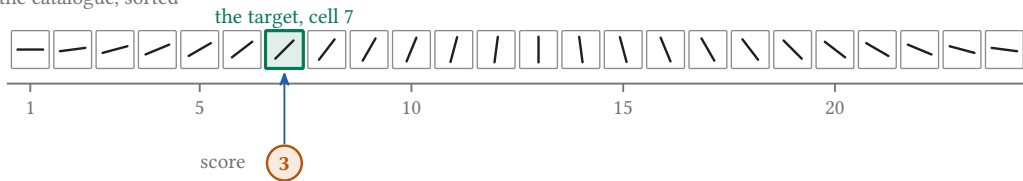
the catalogue, sorted

the target, cell 7



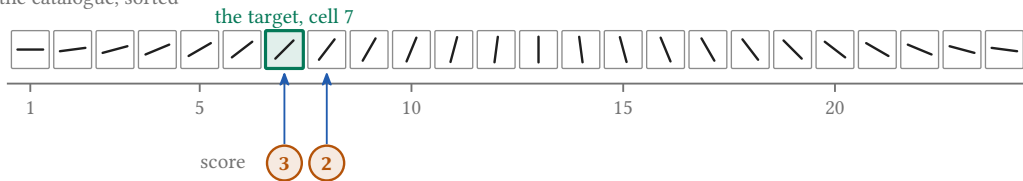
Partial credit

the catalogue, sorted



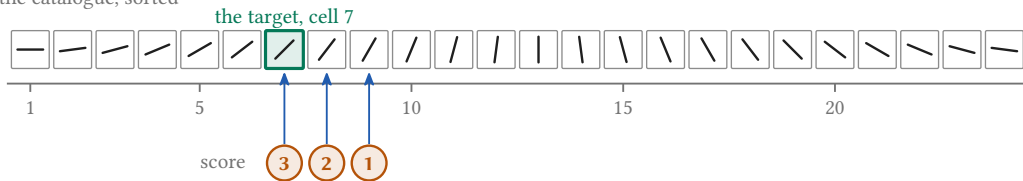
Partial credit

the catalogue, sorted



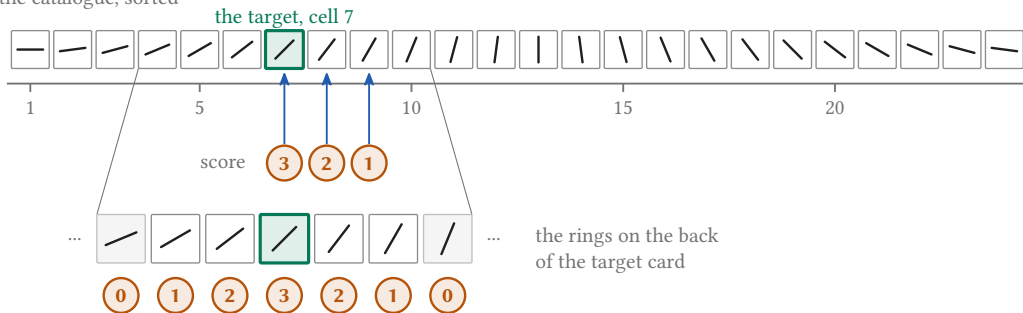
Partial credit

the catalogue, sorted

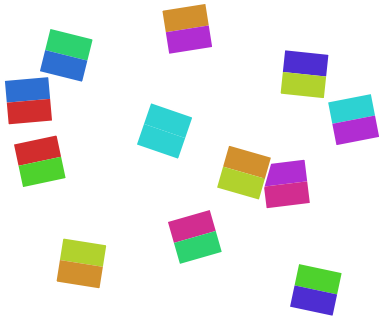


Partial credit

the catalogue, sorted

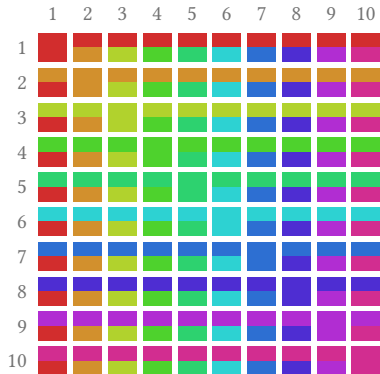


Words are a coarse code



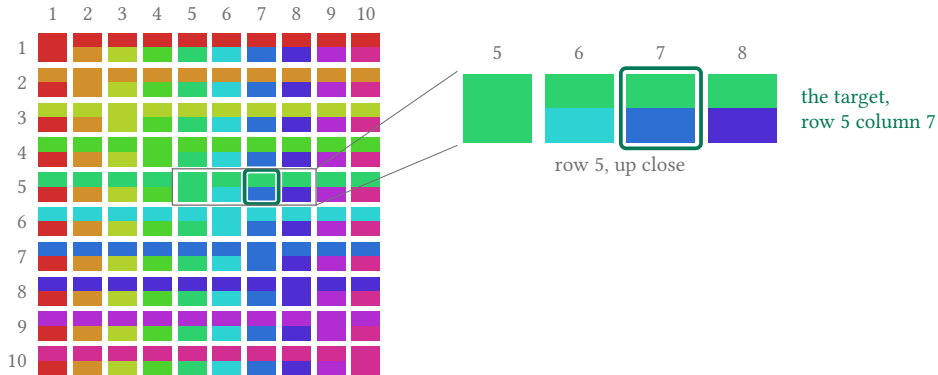
twelve of them, in no order

Words are a coarse code



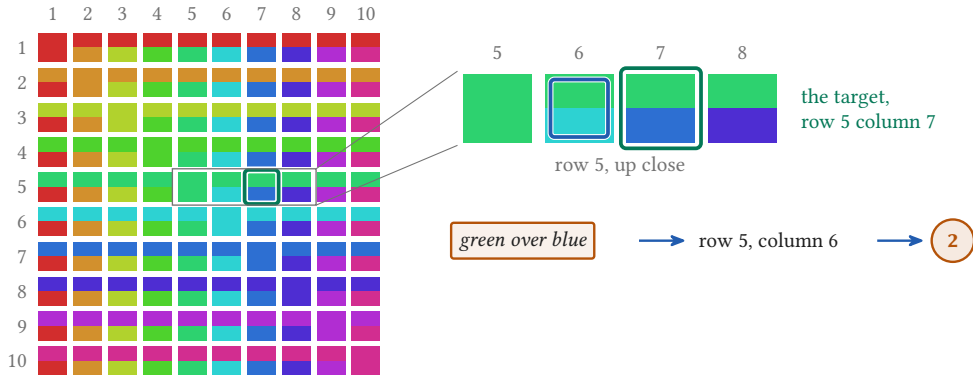
the same flags, sorted on two axes: one hundred, and every one addressable

Words are a coarse code



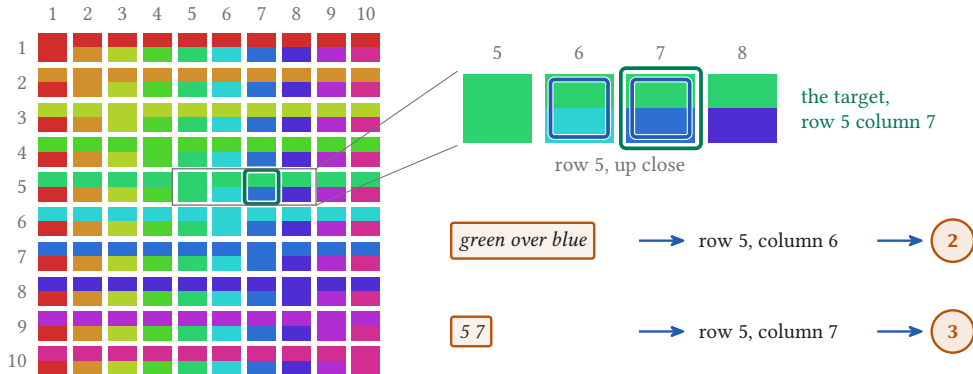
the same flags, sorted on two axes: one hundred, and every one addressable

Words are a coarse code



the same flags, sorted on two axes: one hundred, and every one addressable

Words are a coarse code



the same flags, sorted on two axes: one hundred, and every one addressable

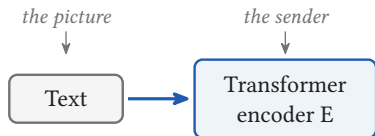
Autoencoders

the picture

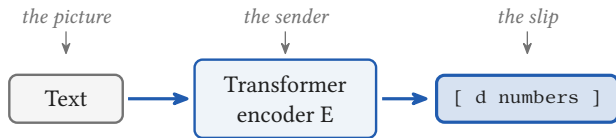


Text

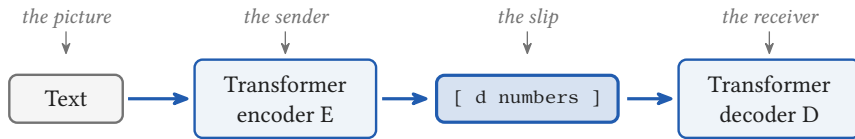
Autoencoders



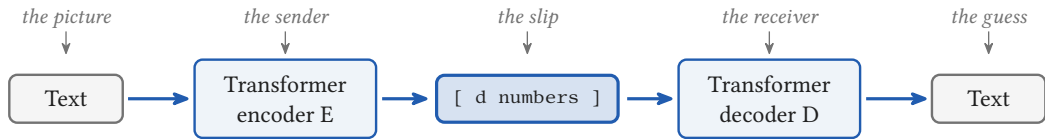
Autoencoders



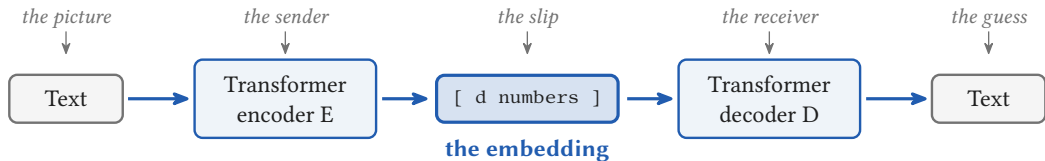
Autoencoders



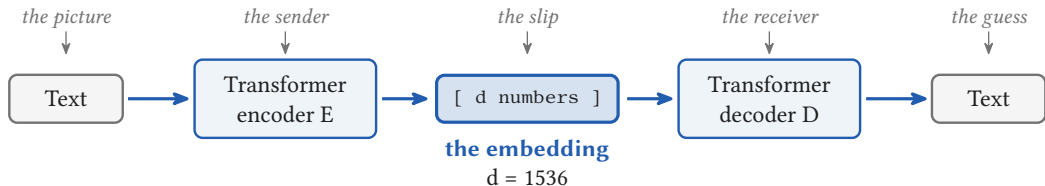
Autoencoders



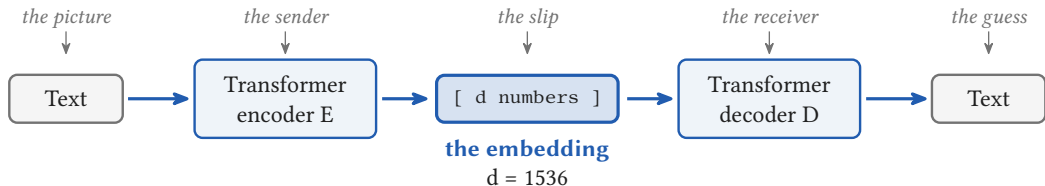
Autoencoders



Autoencoders



Autoencoders



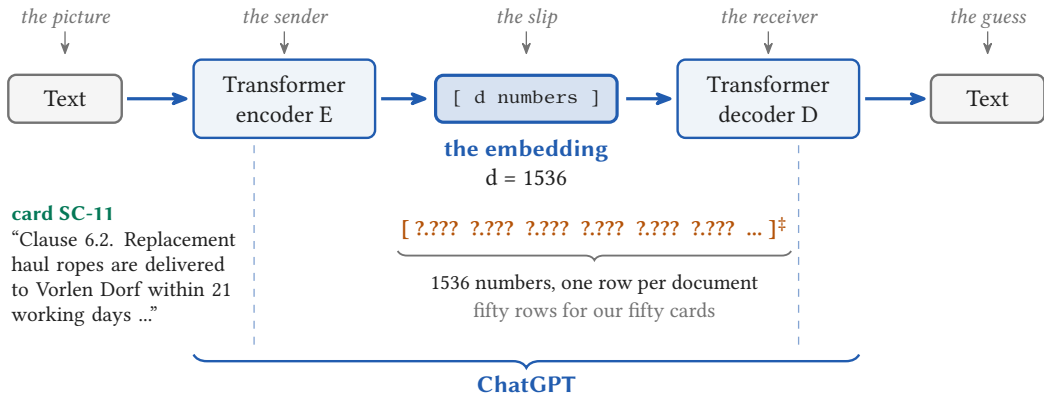
card SC-11

“Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days ...”

[???? ???? ???? ???? ???? ???? ...][‡]

1536 numbers, one row per document
fifty rows for our fifty cards

Autoencoders



How it is trained

- 1 Collect a large collection of texts

How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random

How it is trained

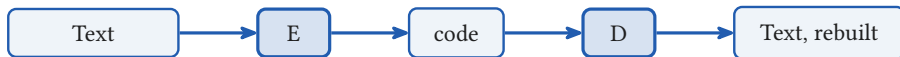
- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score

How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again

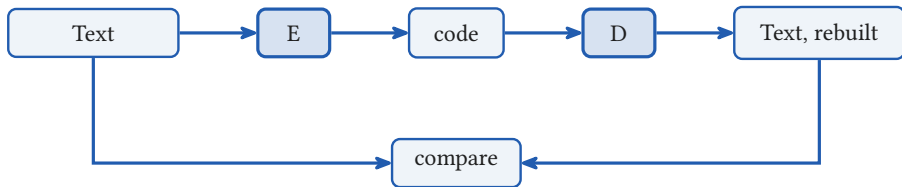
How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again



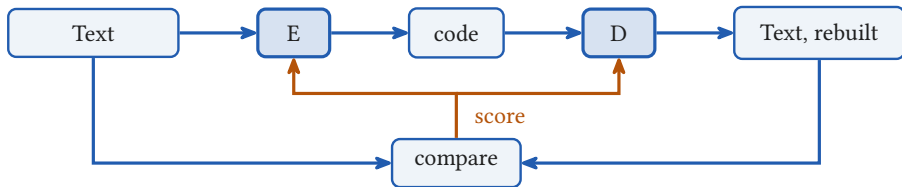
How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again

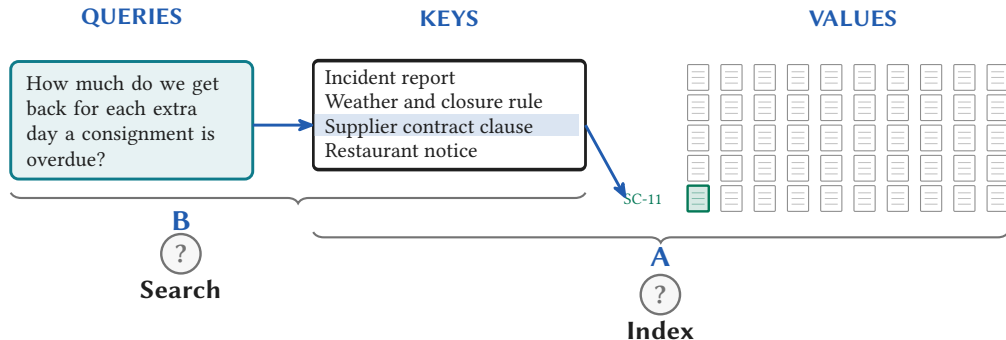


How it is trained

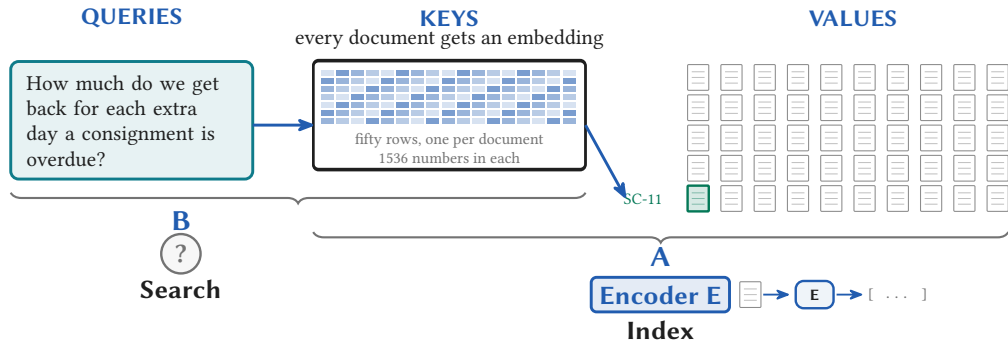
- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again



Queries, keys and values

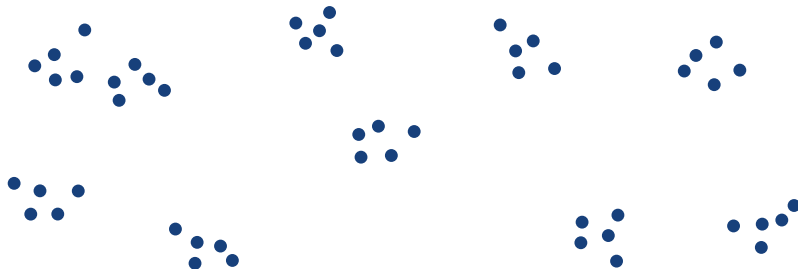


Queries, keys and values



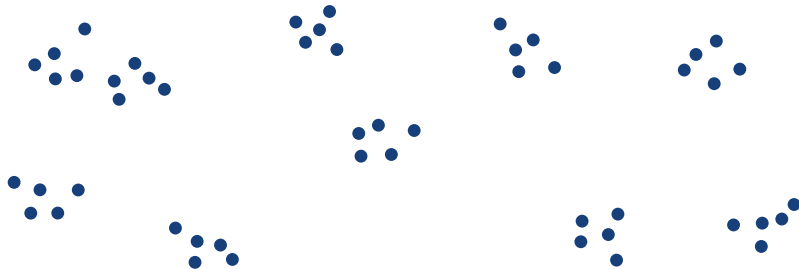
The list of embeddings is the list of keys.

How do we effectively find the relevant keys?



fifty points

How do we effectively find the relevant keys?

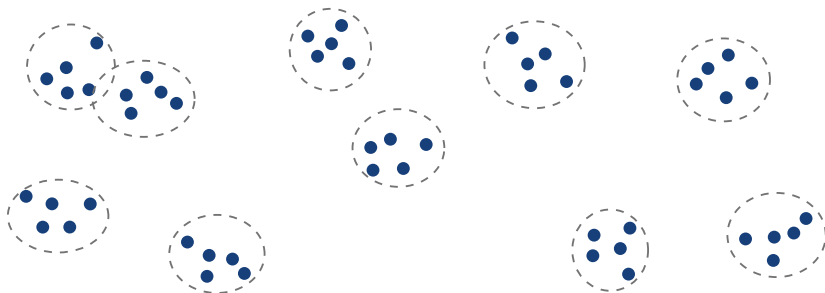


fifty points

illustrative, not measured

A flattened shadow. The clumps are found in all 1536 dimensions, not in this picture.

How do we effectively find the relevant keys?

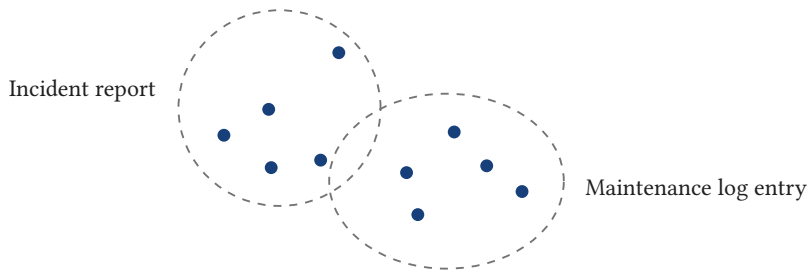


fifty points, ten clumps

illustrative, not measured

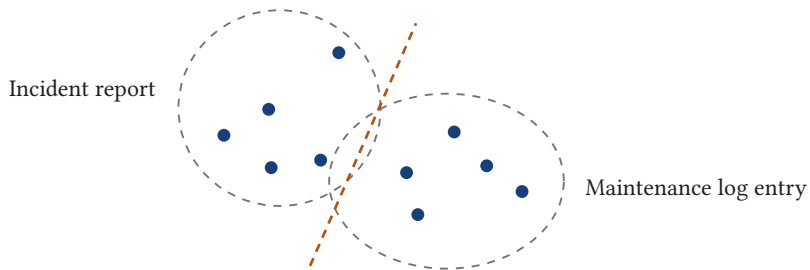
A flattened shadow. The clumps are found in all 1536 dimensions, not in this picture.

Two clumps that touch



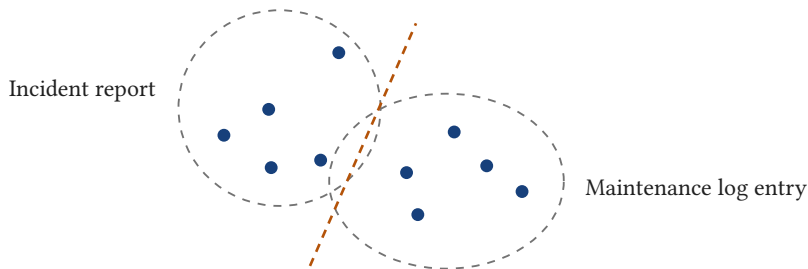
illustrative

Two clumps that touch



illustrative

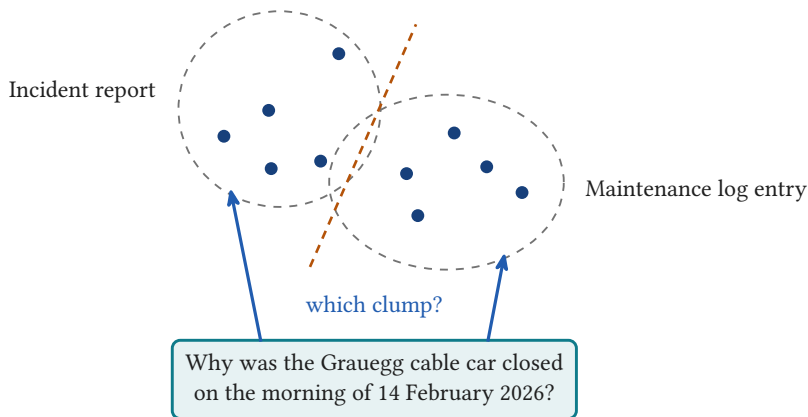
Two clumps that touch



illustrative

Why was the Grauegg cable car closed on the morning of 14 February 2026?

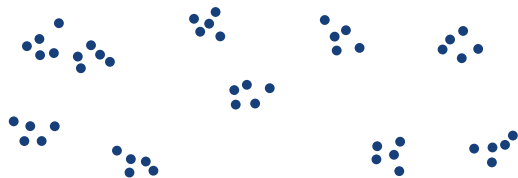
Two clumps that touch



illustrative

Finding the relevant keys

Done once, before any question



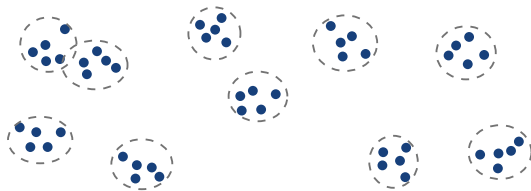
fifty embeddings

illustrative

Finding the relevant keys

Done once, before any question

(k-means, the one most of you have run)

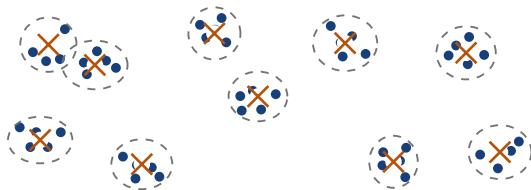


fifty embeddings

illustrative

Done once, before any question

(k-means, the one most of you have run)

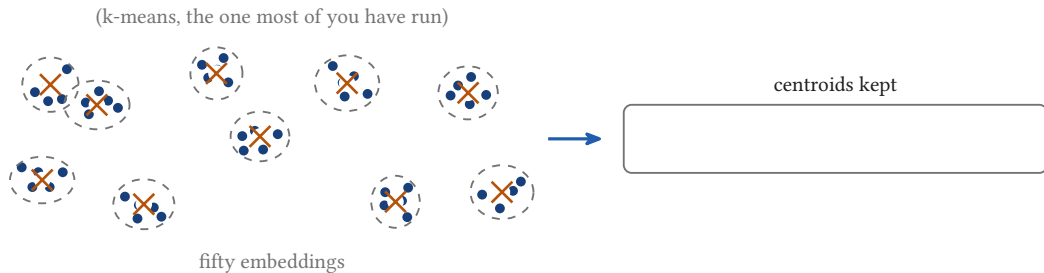


fifty embeddings

illustrative

Finding the relevant keys

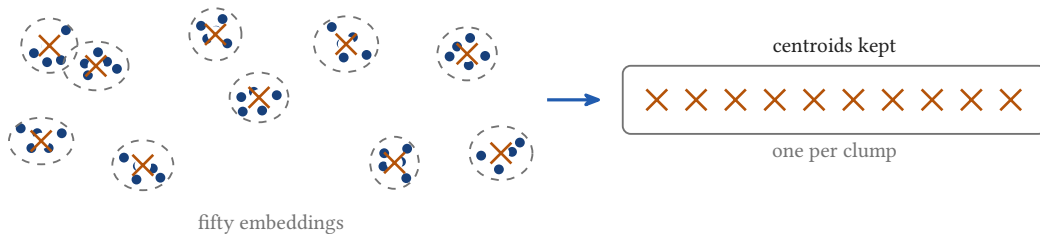
Done once, before any question



illustrative

Done once, before any question

(k-means, the one most of you have run)



illustrative

the pause between rounds, when nobody was asking you anything

At question time

How much do we get back for each extra day
a consignment is overdue?



E

At question time

How much do we get back for each extra day
a consignment is overdue?



E



the query lands
in the same space



At question time

How much do we get back for each extra day
a consignment is overdue?



E



the query lands
in the same space



ten comparisons

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i$$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

0.86

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

five comparisons

Matcher

the question $\mathbf{x} = [0.21 \ -0.04 \ \dots]$

one candidate centroid $\mathbf{y} = [0.19 \ -0.07 \ \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

0.86

illustrative

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

five comparisons



SC-11

Matcher

the question $\mathbf{x} = [0.21 \ -0.04 \ \dots]$

one candidate centroid $\mathbf{y} = [0.19 \ -0.07 \ \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

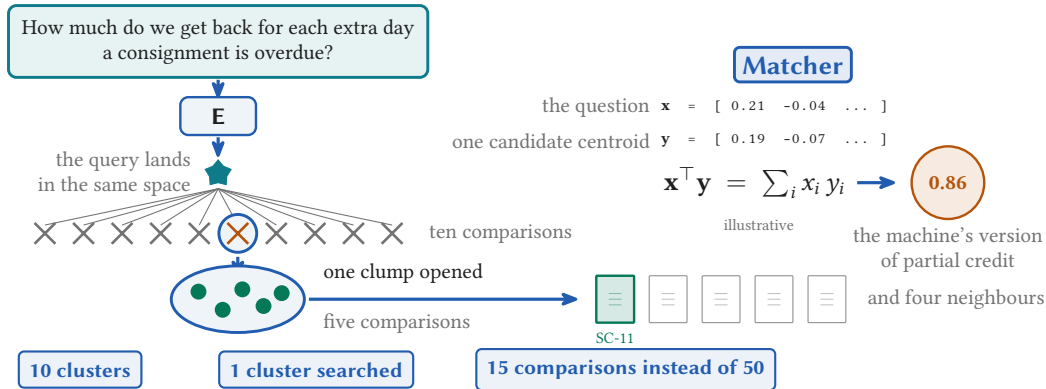
0.86

illustrative

the machine's version
of partial credit
and four neighbours



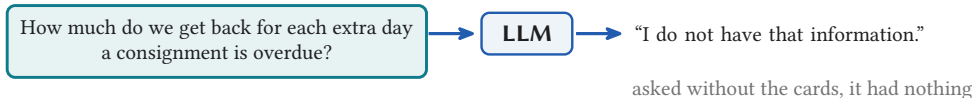
At question time



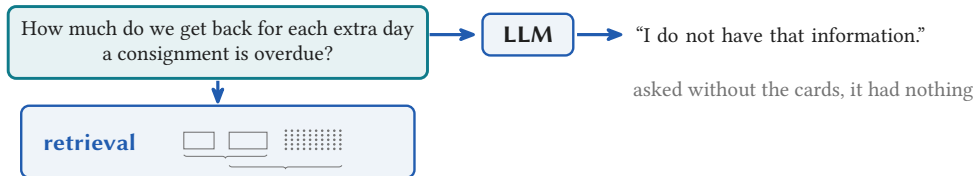
The full system

How much do we get back for each extra day
a consignment is overdue?

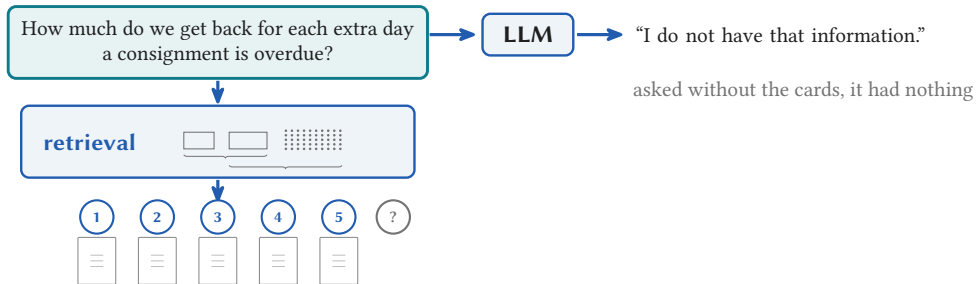
The full system



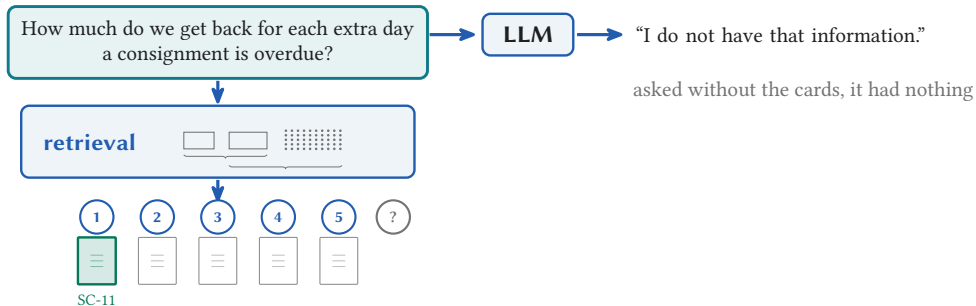
The full system



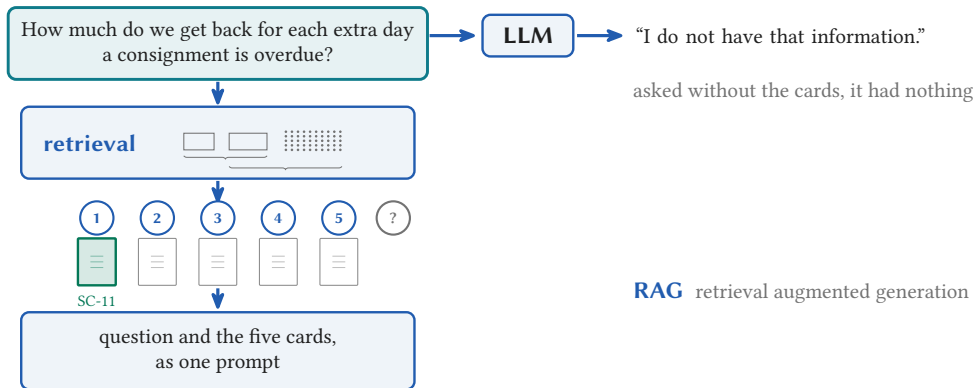
The full system



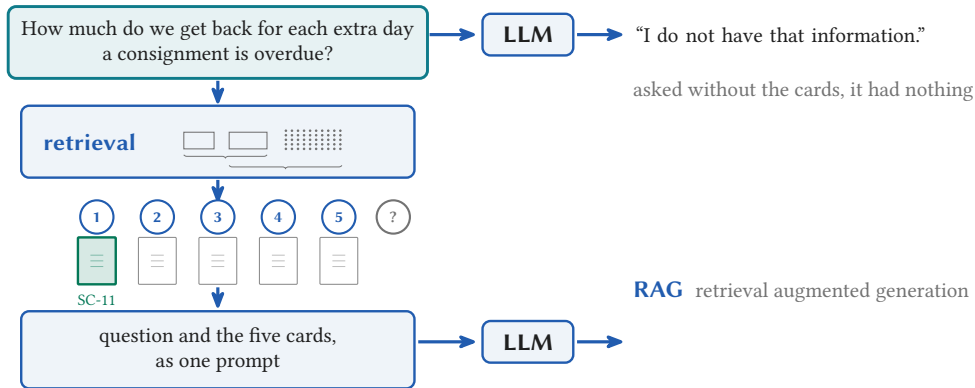
The full system



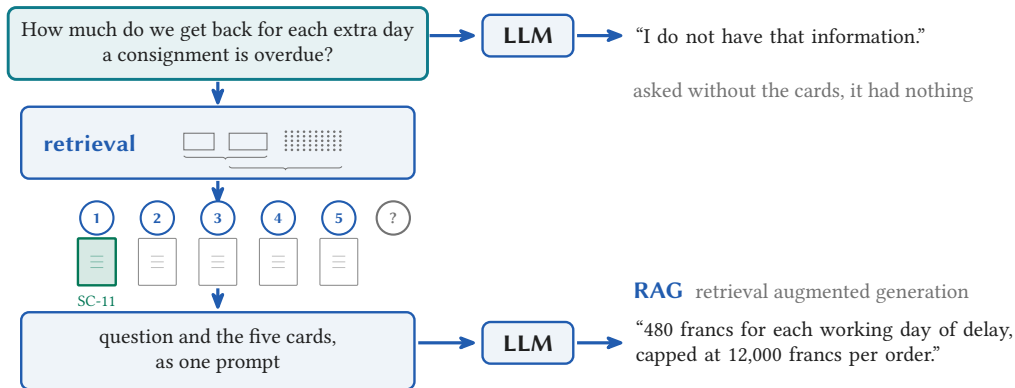
The full system



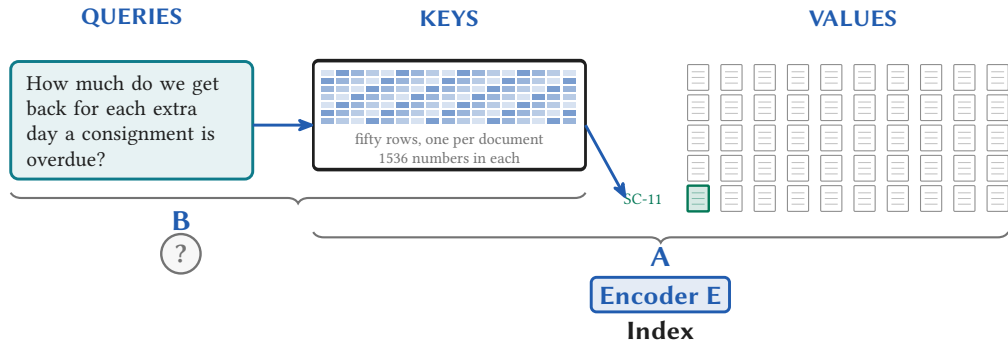
The full system



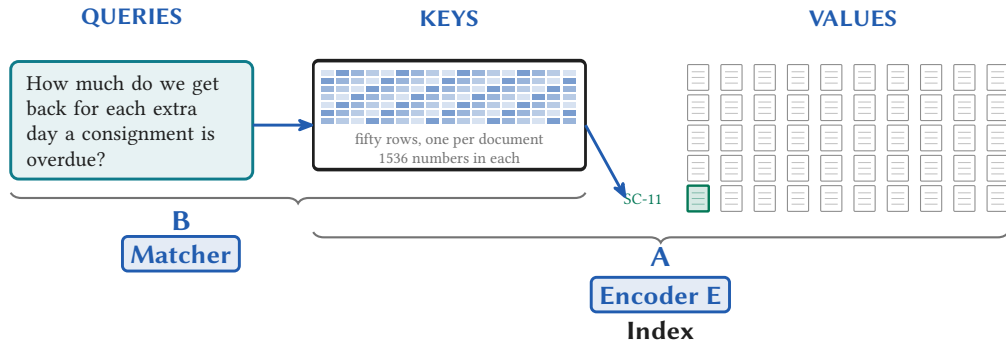
The full system



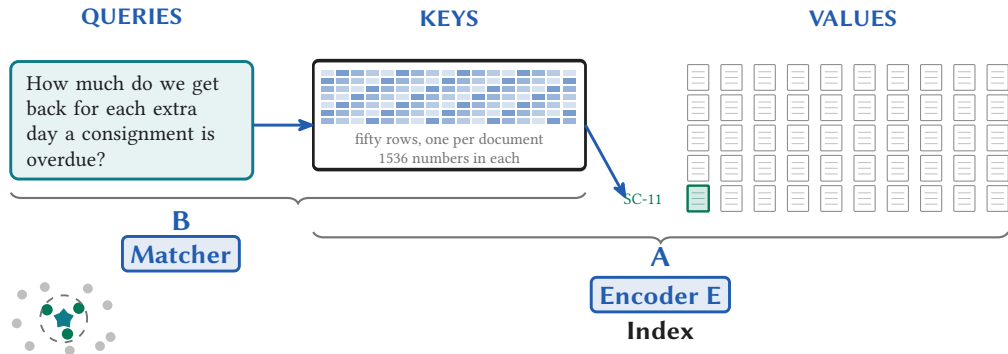
Queries, keys and values



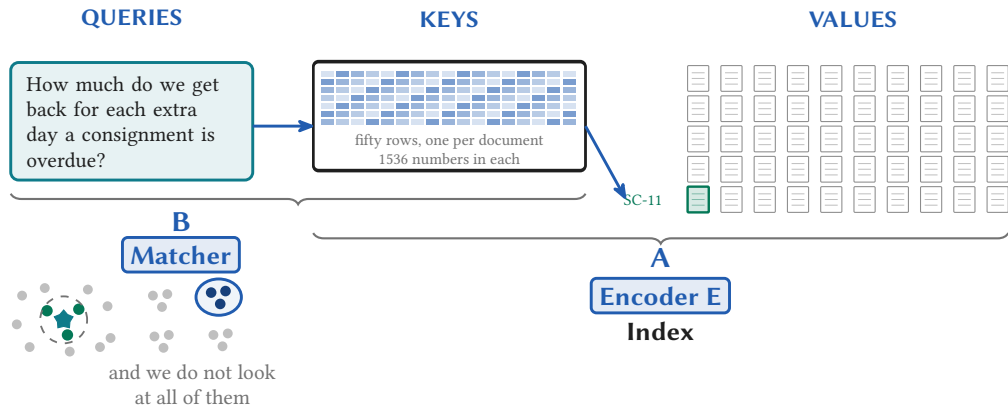
Queries, keys and values



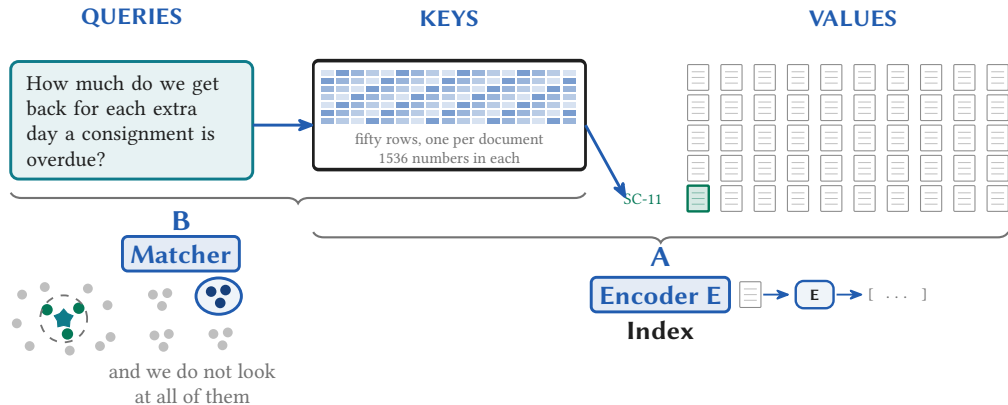
Queries, keys and values



Queries, keys and values



Queries, keys and values



Queries, keys and values

